

Introduction to Machine Learning

Chapter01: Regression

ASST.PROF. PAISIT KHANARSA, PHD

INSTITUTE OF FIELD ROBOTICS (FIBO)

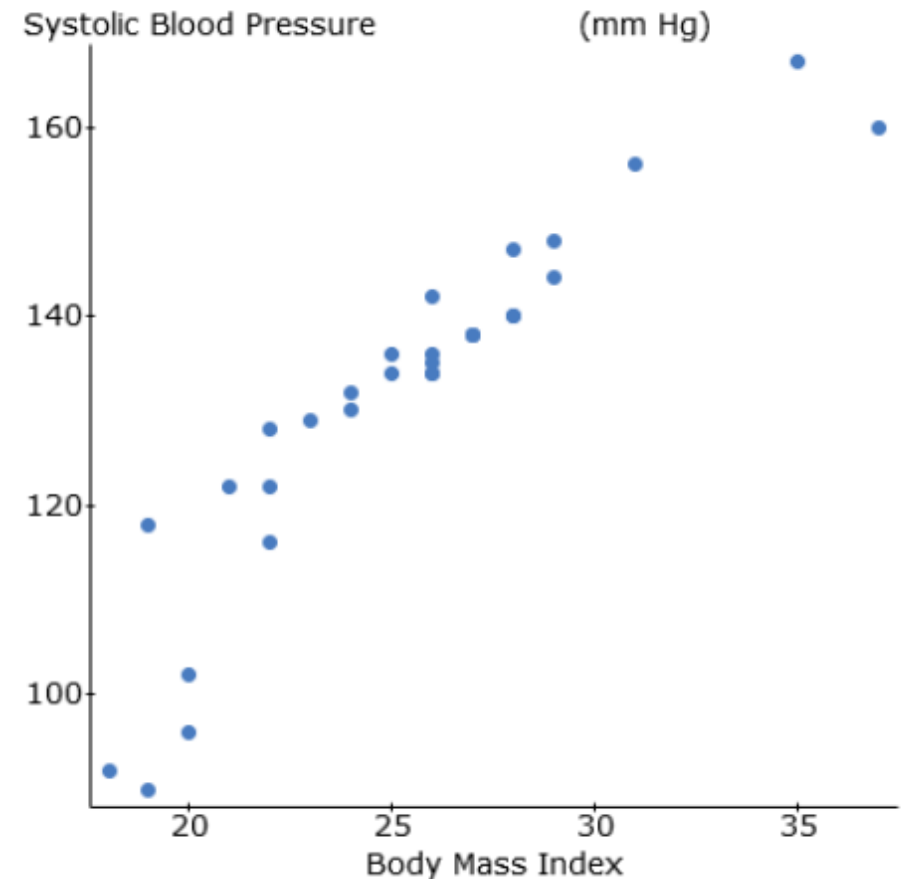
KING MONGKUT'S UNIVERSITY OF TECHNOLOGY THONBURI (KMUTT)

Outlines

- Motivation
- Regression Task
- Linear Least Square Regression Model
- Over Fitting
- Some Examples
- Regression Analysis
- Basic Data Representation in Machine Learning Process

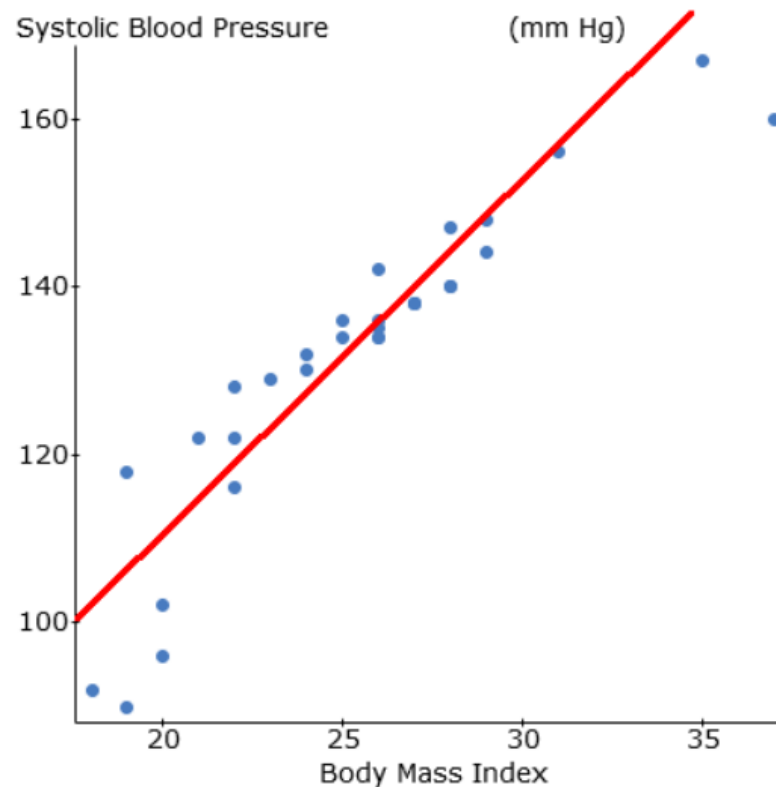
Motivation

- Suppose we have records of patients BMI and blood pressures
- And we would like to estimate relationship between BMI and blood pressure, so that we could
 - predict blood pressure if we know BMI



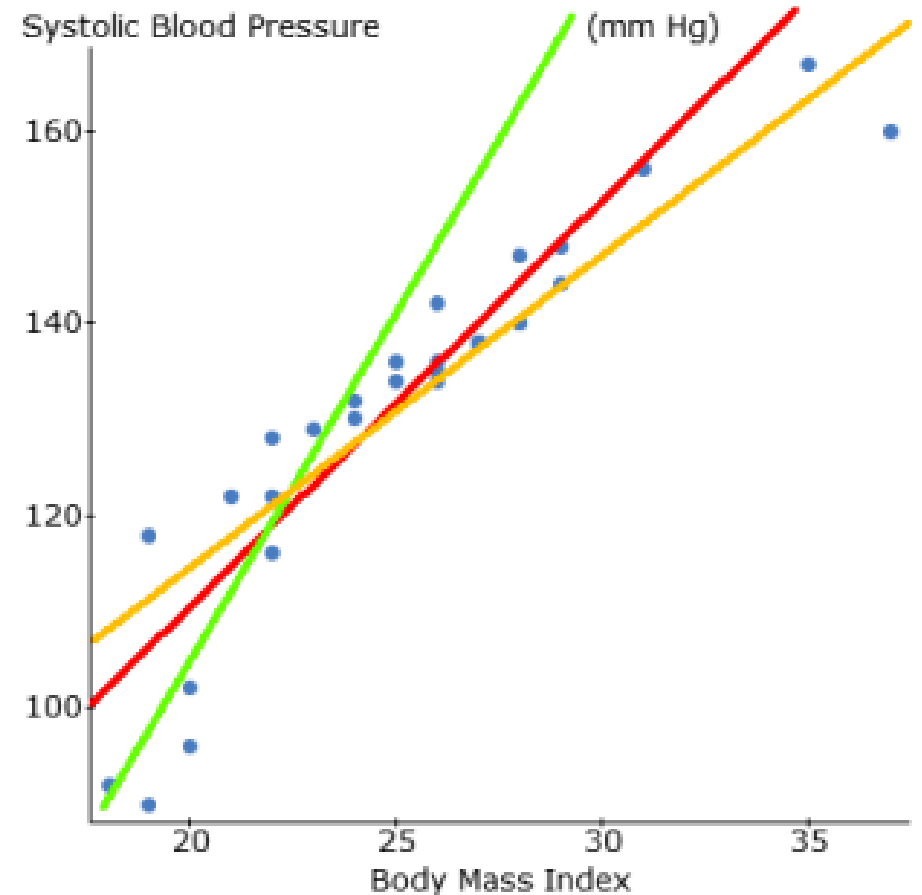
Motivation

- Assuming that the relationship between feature and target is linear, we could fit a linear function (a straight line) to the data.



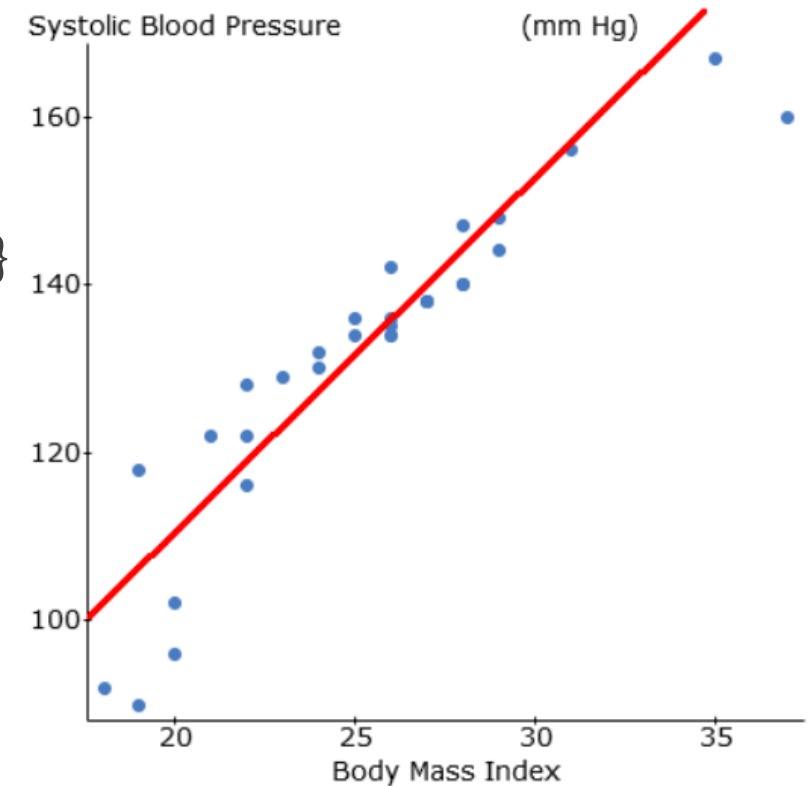
Motivation

- Assuming that the relationship between feature and target is linear, we could fit a linear function (a straight line) to the data.
- There are many possible lines, which one is the best fit ?
- How to measure goodness of fit ?



Regression task

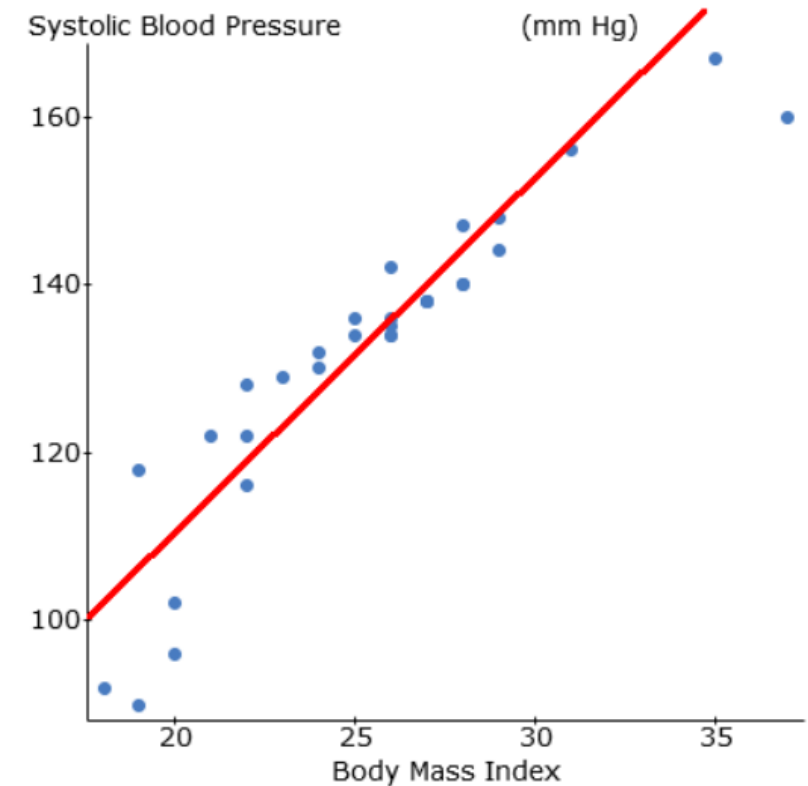
- Regression analysis is a study of methods for estimating the relationship between independent variable(s)/input feature(s) and dependent variable/target
- Given a dataset $\{(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)\}$
- $f(x_i) = a_0 + a_1 x_i$
- General $\{(x_1^1, \dots, x_1^M, y_1), (x_2^1, \dots, x_2^M, y_2), \dots (x_n^1, \dots, x_n^M, y_n)\}$
- $f(\mathbf{x}) = a_0 + a_1 x_i^1 + a_2 x_i^2 + \dots a_M x_i^M = \sum_k^M a_k x_i^k = \mathbf{a}^T \mathbf{x}$
- where $\mathbf{x} = [1, x_i^1, \dots, x_i^M]^T$, $\mathbf{a} = [a_0, a_1, \dots, a_M]^T$
- n data points and M features
- How to measure goodness of fit ?



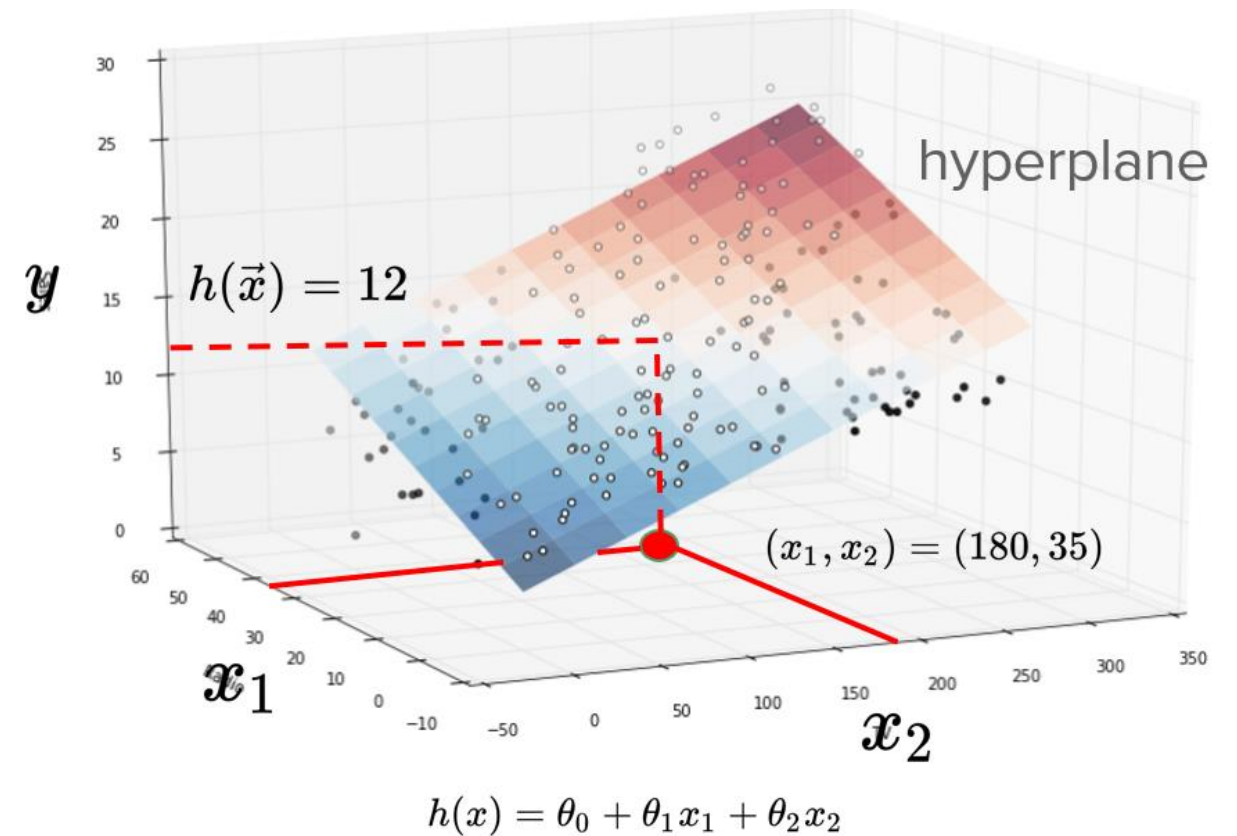
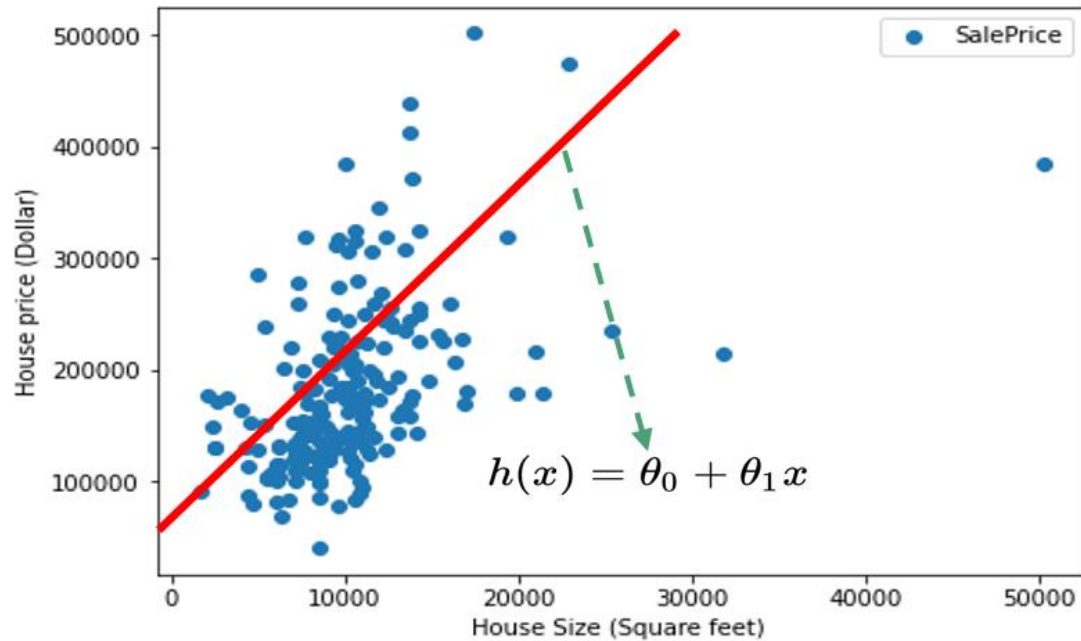
Regression task

- Regression analysis is a study of methods for estimating the relationship between independent variable(s)/input feature(s) and dependent variable/target
- Given a dataset $\{(x_1, y_1), (x_2, y_2), \dots (x_n, y_n)\}$
- How to measure goodness of fit ?
- Cost function (loss function) is a measure of whether a model is a good fit to the data.
- For example, a famous cost function is called 'squared error' function (Sum of Squared Residuals (SSR)) that takes the difference between what you predict and the actual data value and square it

$$\sum_{i=1}^n (y_i - f(\mathbf{x}_i))^2$$

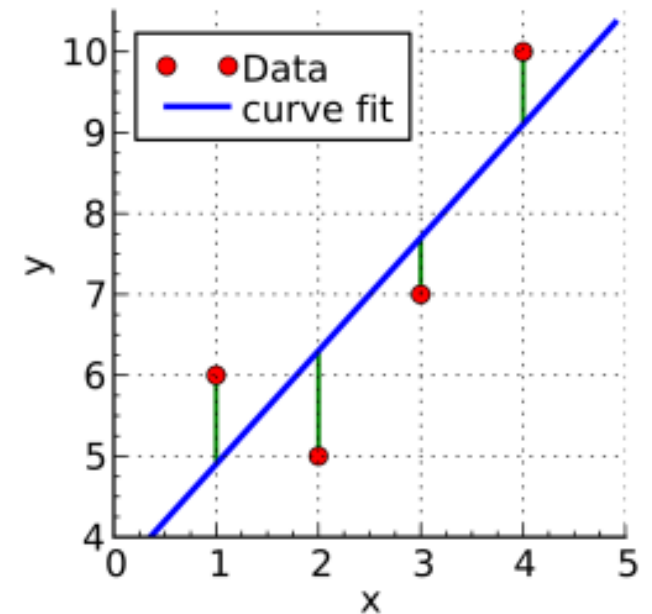


Regression task



Linear Least Square Regression Model

- Recall that in linear regression problem we are given a set of data $(x_i, y_i), i = 0, \dots, N$ (red dots)
- And we want to find a linear function $f(x)$ that best describes the data
- In most of the cases, $f(x)$ cannot perfectly describe the data
- There is almost always mismatch between y_i (the true value) and the value given by the model $f(x_i)$
- The magnitude of the mismatch is length of green line



Linear Least Square Regression Model

- The error of the estimation of each y_i is given by

$$\begin{aligned}\varepsilon_i &= y_i - f(x_i) \\ &= y_i - ax_i\end{aligned}$$

- Since ε_i can be either positive or negative, their values might cancel out
- So we square the error terms and get

$$\varepsilon_i = (y_i - ax_i)^2$$

- Summing all the error terms we have

$$\varepsilon = \sum_{i=0}^N (y_i - ax_i)^2$$

- a is a coefficient vector of the linear model and N =number of data points

Linear Least Square Regression Model

- We can write the sum of squared errors in matrix form

$$\varepsilon = \|Xa - Y\|_2^2$$

- X is a data matrix N by M (N =number of data points M = data dimension)
- Y is matrix of responses and a is a coefficient vector of the linear model
- Examples: given $\{(3, 14), (2, 13), (5, 19)\}$

- 1 dimension

$$X = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}, a = \begin{bmatrix} a_1 \end{bmatrix}, Y = \begin{bmatrix} 14 \\ 13 \\ 19 \end{bmatrix}$$

- 2 dimensions

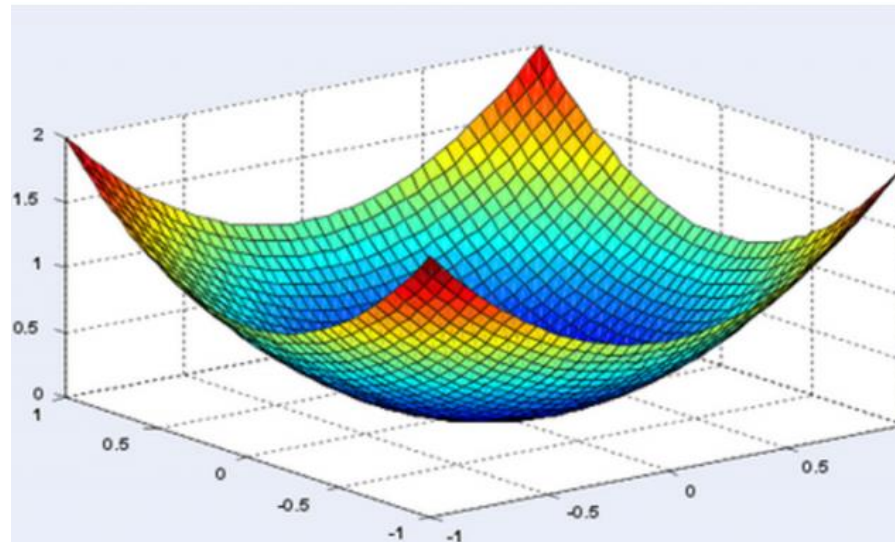
$$X = \begin{bmatrix} 3 & 2 \\ 2 & 1 \\ 5 & 4 \end{bmatrix}, a = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, Y = \begin{bmatrix} 3.3 \\ 1.5 \\ 7.9 \end{bmatrix}$$

Linear Least Square Regression Model

Goal: Find a vector a which when multiplied to all the data gives the smallest sum of squared errors (the objective function)

$$L = \min_a \|Xa - Y\|_2^2$$

The function is a convex function with a global minimum



Linear Least Square Regression Model

A point where the minimum is, is the stationary point (slope/derivative) is zero.

First we find the derivative of the objective function w.r.t a

$$\frac{\partial L}{\partial a} = \frac{\partial \|Xa - Y\|_2^2}{\partial a} = 2X^T(Xa - Y).$$

Next find a which gives zero slope

$$X^T(Xa - Y) = 0$$

$$a = (X^T X)^{-1} X^T Y.$$

Learning Algorithm: Gradient Descent

- Have some function $L(w,b)$
- Want to minimize $L(w,b)$ w.r.t w,b

∇f = direction of steepest ascent

- **Outline:**

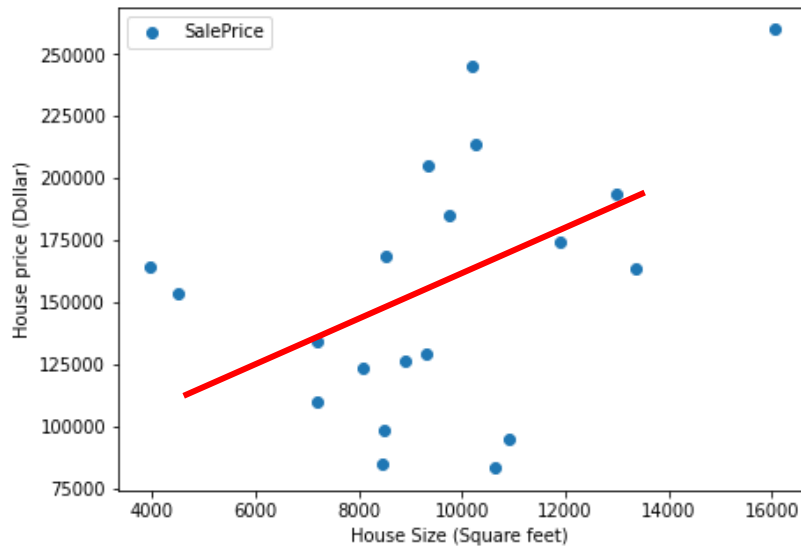
- Start with some w,b
- Keep changing w,b to reduce $L(w,b)$
 - finding Gradient vector of $L(w,b)$
 - Gradient is a vector of partial derivatives that points in the direction of the steepest increase of a function.
 - Walk in the opposite direction of the gradient.
- Until we settle at or near a minimum

Overfitting

In machine learning, we need to minimize cost function, however sometimes, we have the opposite problem when the cost function is too low.

Underfit

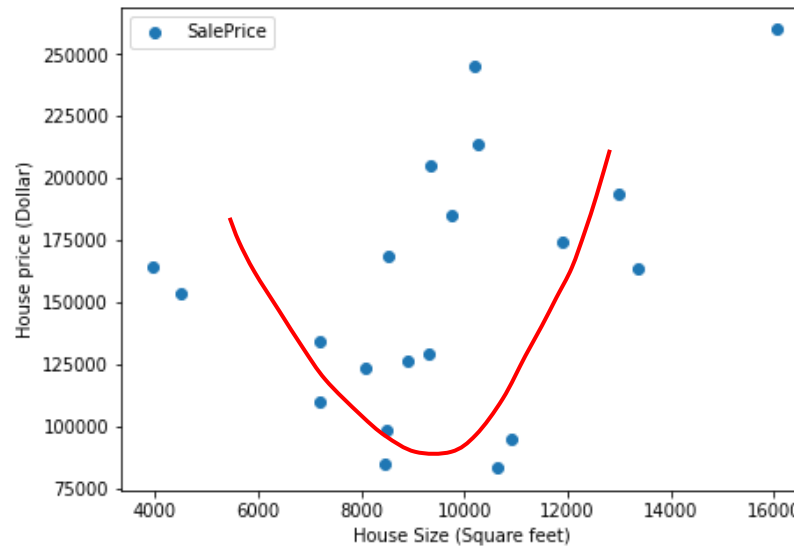
$Cost = 0.5$



$$h(x) = \theta_0 + \theta_1 x$$

Right

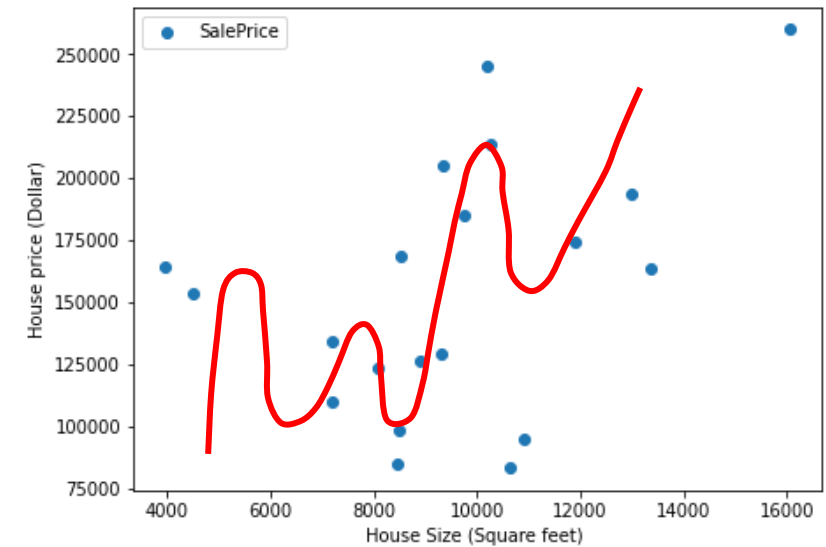
$Cost = 0.04$



$$h(x) = \theta_0 + \theta_1 x + \theta_2 x^2$$

Overfit

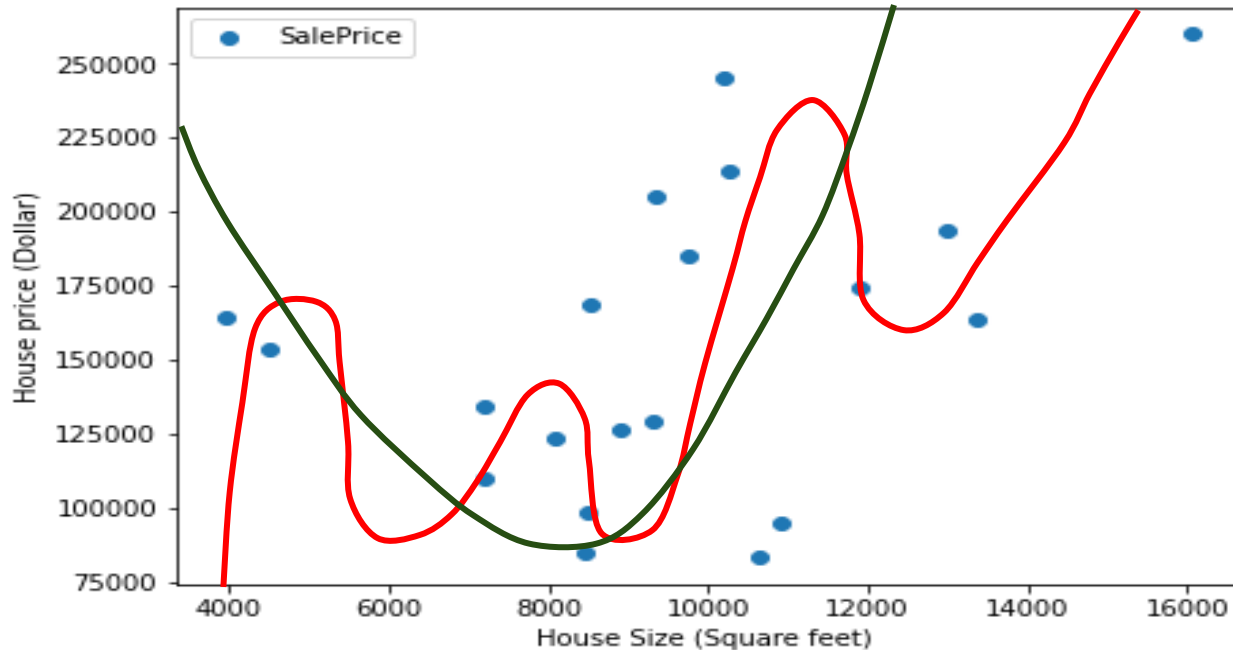
$Cost = 0.01$



$$h(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_9 x^9$$

Overfitting

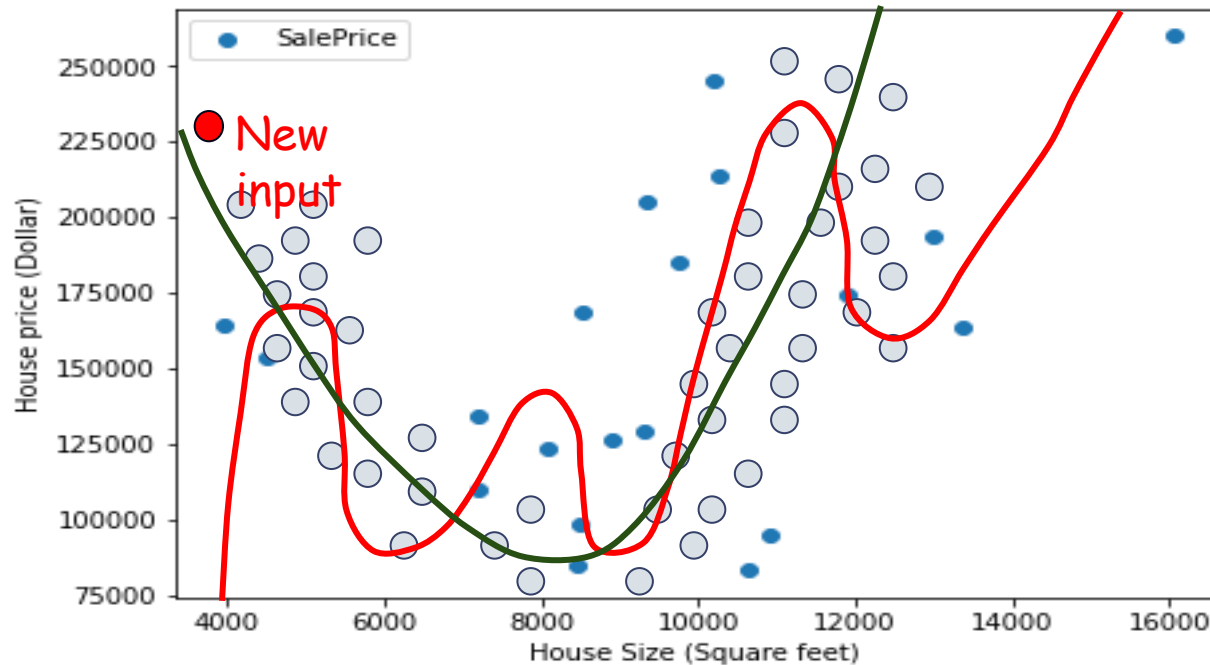
In machine learning, we need to minimize cost function, however sometimes, we have the opposite problem when the cost function is too low.



- ❖ Training data: good predictions
- ❖ Data outside training set: bad predictions
- ❖ Overfitting
 - ❖ Signal to noise
 - ❖ Memorize noise

Overfitting

In machine learning, we need to minimize cost function, however sometimes, we have the opposite problem when the cost function is too low.



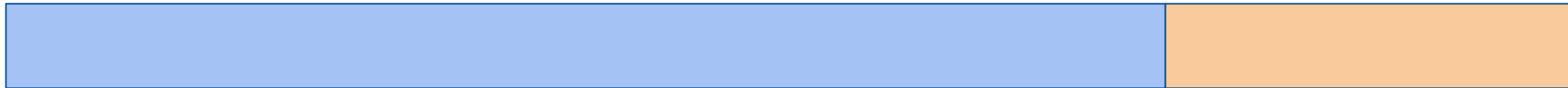
Fact about over fitting

- ❖ The more parameters you have, the more likely your model will overfit.
- ❖ The more data you have (compared to parameters) the less likely your model will overfit, because noise will be more likely to average out.

Avoid overfitting

Cross-validate your models: train models with one set of data (training set) and test them with another set of data (test set).

Cross validation



70% of data are used
for training

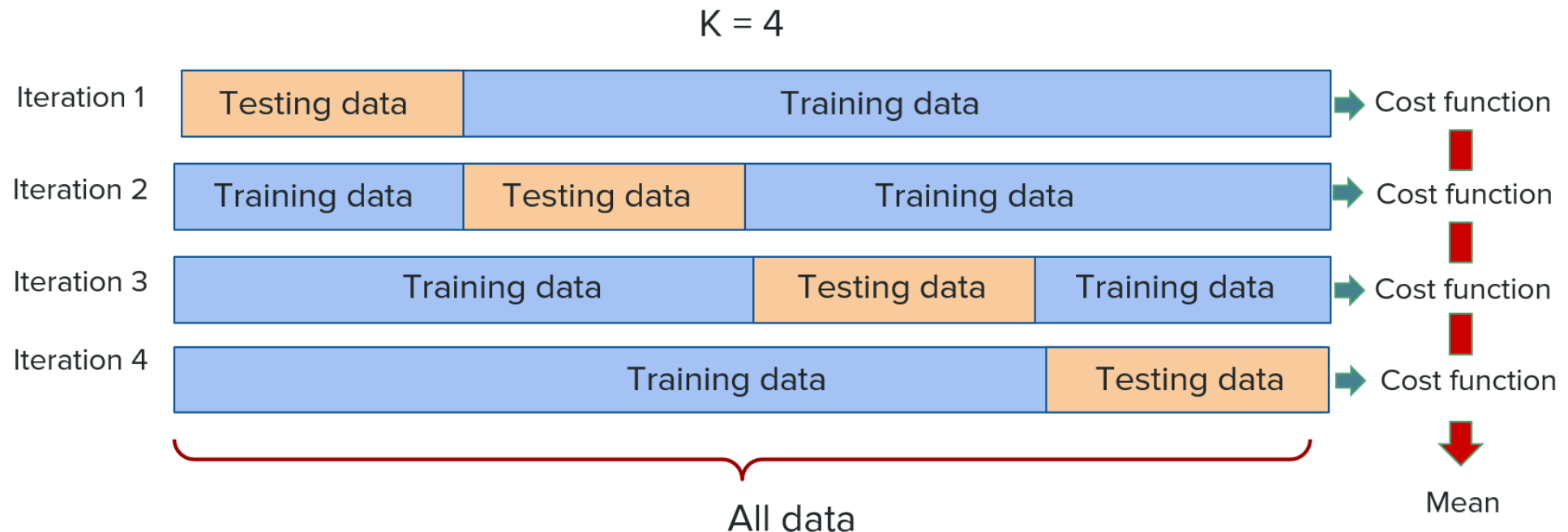
30% of data are used
for testing

Overfitting model

- ❑ Train data → low cost but test data → high cost

Avoid overfitting

- F-fold cross validation: divide data into K chunks.
- Fit the model K times, each time use one chunk as test data, the rest as training data.



Example: BMI and BP data

- x_i is a BMI of one patient (since we have one feature (BMI) our data is in 1-D)
- y_i is his blood pressure
- Data is of the form $\{(20, 102), (23, 119), \dots, (35, 170)\}$
- Assuming linear relationship we want to find a linear function
$$f(x) = ax + b$$
- a and b are model's parameters that need to be tuned
- so that $\text{SSR} = \varepsilon = \sum_{i=0}^N (y_i - f(x_i))^2$ is minimized
- This is a supervised-learning because we use targets (answers) to help finding the best model's parameters.

Example: Secondhand car pricing

- In many problems, we may have several features. $x = [x_1, x_2]$
 - x_1 = car's CC, x_2 = mileage
 - y is its second hand price
- Data is of the form $\{([1600, 150000], 200000), ([2000, 50000], 450000), \dots, \}$
- Assuming linear relationship we want to find a linear function
$$f(\mathbf{x}) = ax_1 + bx_2 + c$$
- a, b and c are model's parameters that need to be tuned

Regression Analysis

Mean squared error

$$\text{MSE} = \frac{1}{n} \sum_{t=1}^n e_t^2$$

Root mean squared error

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n e_t^2}$$

Mean absolute error

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |e_t|$$

Mean absolute percentage error

$$\text{MAPE} = \frac{100\%}{n} \sum_{t=1}^n \left| \frac{e_t}{y_t} \right|$$

where $e_t = (y_t - \hat{y}_t)$

R-squared in Regression Analysis

R-squared (R^2), or the **Coefficient of Determination**, is a statistical measure that assesses the proportion of variance in the dependent variable that is predictable from the independent variables.

It provides an indication of how well the independent variables explain the variation in the dependent variable.

- Formula for R-squared:

$$R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

- SS_{res} (Residual Sum of Squares): The sum of the squared differences between the actual values and the predicted values.
- SS_{tot} (Total Sum of Squares): The sum of the squared differences between the actual values and the mean of the actual values.

$$SS_{res} = \sum (y_i - \hat{y}_i)^2$$

$$SS_{tot} = \sum (y_i - \bar{y})^2$$

- y_i = Actual value of the dependent variable
- $\hat{y}_i$ = Predicted value from the model
- $\bar{y}$ = Mean of the actual values

R-squared in Regression Analysis

Interpretation of R-squared

Range of R-squared:

- $R^2=1$: The model explains all the variability of the response data around its mean.
- $R^2=0$: The model explains none of the variability of the response data around its mean.
- $R^2<0$: The model performs worse than a horizontal line representing the mean of the response variable.

Example Interpretation:

- If $R^2=0.85$, it means that 85% of the variability in the dependent variable is explained by the independent variables, leaving 15% unexplained.

Linearity check

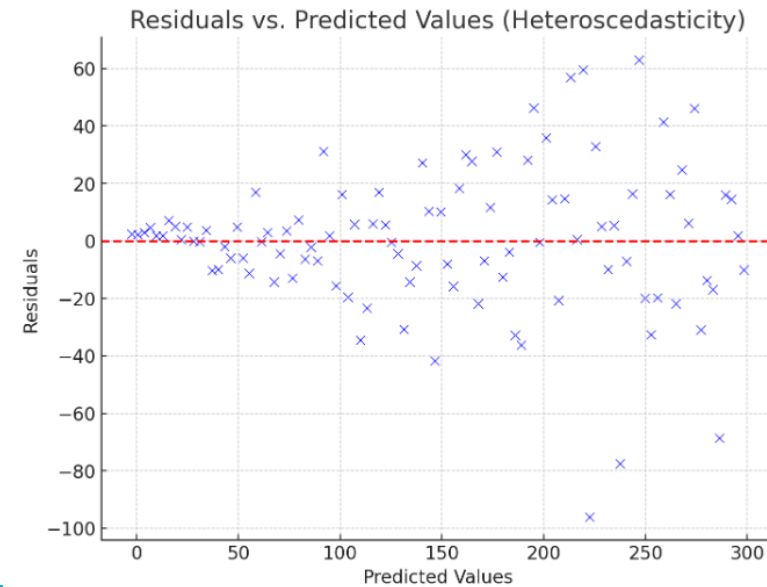
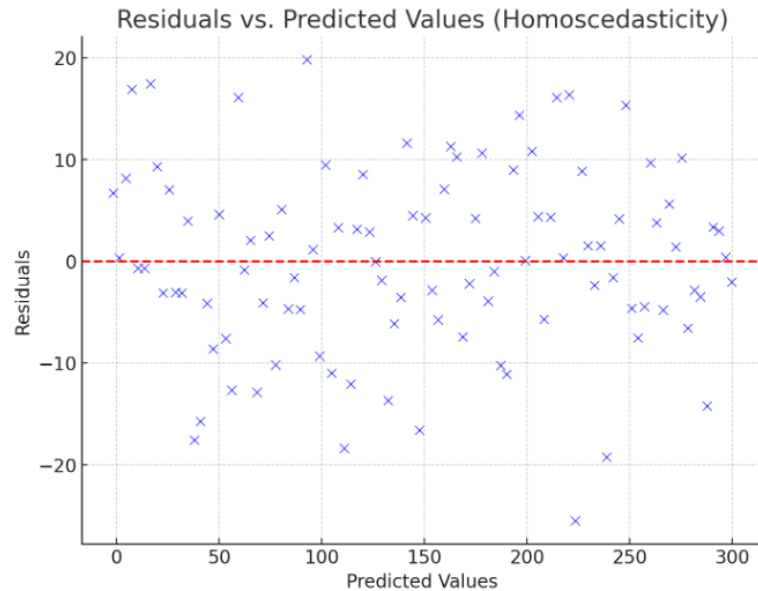
There are 4+ key assumptions on data that can be used to check if linear regression is appropriate for the data.

- Linear Relationship between the features and target
- Little or no Multicollinearity between the features
- Normal distribution of error terms (residual) (with mean zero and variance σ^2)
- No High Leverage Points or Influential Outliers
- No auto-correlation
 - "There is no temporal correlation, meaning that the values occurring at one time may have no relationship with the values occurring at a previous time."
- Homoscedasticity
 - Homoscedasticity refers to the situation where the variance of the residuals (or errors from the predicted line) is consistent across the range of predicted values or independent variable values.

Linearity check

There are 4+ key assumptions on data that can be used to check if linear regression is appropriate for the data.

- Homoscedasticity
 - Homoscedasticity refers to the situation where the variance of the residuals (or errors from the predicted line) is consistent across the range of predicted values or independent variable values.



Basic Data Representation in Machine Learning Process

Data Matrix

Data Representation

Example: Cosmic Dataset

	name X_1	id X_2	align X_3	eye X_4	hair X_5	gender X_6	alive X_7	appearances X_8	first_appear X_9	publisher X_{10}
x_1	Spider-Man (Peter Parker)	Secret	Good	Hazel Eyes	Brown Hair	Male	Living Characters	4043	Aug-62	marvel
x_2	Captain America (Steven Rogers)	Public	Good	Blue Eyes	White Hair	Male	Living Characters	3360	Mar-41	marvel
	...	...	...	...	...	...	...	...	...	...
x_n	Natalia Romanova (Earth-616)	Public	Good	Green Eyes	Red Hair	Female	Living Characters	1050	Apr-64	marvel

Data Matrix

Data Representation

$$\text{Dataset } D = \begin{matrix} & \overbrace{\begin{matrix} X_1 & X_2 & \cdots & X_d \end{matrix}}^{\text{Attributes, Variable, Features, Field,...}} \\ \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1d} \\ x_{21} & x_{22} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nd} \end{pmatrix} & \begin{matrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{matrix} \end{matrix} \left. \vphantom{\begin{matrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{matrix}} \right\} \begin{matrix} \text{Entities, Instances,} \\ \text{Examples, Records,} \\ \text{Points, Feature-vectors,} \end{matrix}$$

$\mathbf{x}_i$ denotes the i th row which is a d -tuple given as

$$\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{id})$$

X_j denotes the j th column which is a n -tuple given as

$$X_j = (x_{1j}, x_{2j}, \dots, x_{nj})$$

Data Matrix

Data Representation

Example: Cosmic Dataset

	name X_1	id X_2	align X_3	eye X_4	hair X_5	gender X_6	alive X_7	appearances X_8	first_appear X_9	publisher X_{10}
x_1	Spider-Man (Peter Parker)	Secret	Good	Hazel Eyes	Brown Hair	Male	Living Characters	4043	Aug-62	marvel
x_2	Captain America (Steven Rogers)	Public	Good	Blue Eyes	White Hair	Male	Living Characters	3360	Mar-41	marvel
	...	...	...	...	...	...	...	...	...	...
x_n	Natalia Romanova (Earth-616)	Public	Good	Green Eyes	Red Hair	Female	Living Characters	1050	Apr-64	marvel

We can write an example x_2 as

$x_2 = (\text{Captain America (Steven Rogers), Public, Good, Blue Eyes, White Hair, Male, Living Characters, 3360, Mar - 41, marvel})$

Types of Data

Data Representation



Quantitative Data

- This data can be described using **numbers**.
- **Basic mathematical procedures** are possible on the set.



Qualitative Data

- This data cannot be described using numbers and basic mathematics.
- This data is generally described using natural **categories and language**.

Attributes

Data Representation



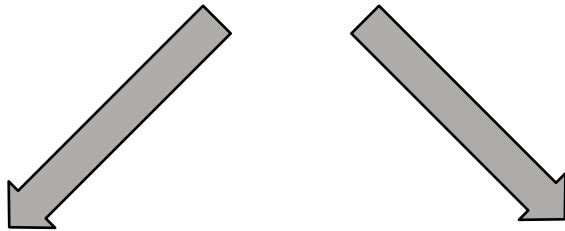
Numeric Attributes - Quantitative

- One that has a real-valued or integer-valued domain.
- Such as age, height, grade, frequency, etc.



Categorical Attributes

- One that has a set-valued domain composed of a set of symbols.
- Such as Gender = {M,F},
Education = {High School, BS, MS, PhD},
etc.



Discrete

- Take on a finite or countably infinite set
- Such as integer, grade, number of object, etc.

Continuous

- Take on any real value
- Such as height, weight, size, etc.

Attributes

Data Representation



Categorical Attributes

Nominal

- Attribute values in the domain are unordered.
- Can only equality (=) compare.
- Such as gender, type of hair, etc.

Ordinal

- Attribute values are ordered.
- Can both equality (=) and inequality (<, >) compare.
- Such as education, feel (unhappy, OK, happy), etc.

Attributes

Data Representation



Numeric Attributes

Interval-scaled

- Can compute only differences (addition or subtraction)
- For example, temperature measured in °C or °F.
 - If it is 20 °C on one day and 10 °C on previous day
 - We **can** talk about a temperature drop of 10°C.
 - We **cannot** say that it is twice as cold as the previous day.

Ratio-scaled

- Can compute both differences and ratio between values,
- For example, age.
 - If Jone is 20 years old and Jim is 10 years old.
 - We **can** say that Jone older than Jim with 10 years.
 - We **can** say that Jone is twice as old as Jim.

Attributes

Data Representation

Summary of data types and scale measures

Provides	Nominal	Ordinal	Interval-scaled	Ratio-scaled
The order of values is known		/	/	/
“Count,” aka “Frequency of Distribution”	/	/	/	/
Mode	/	/	/	/
Median		/	/	/
Mean			/	/
Can quantify the difference between each values			/	/
Can add or subtract values			/	/
Can multiple and divide values				/
Has “true zero”				/

Encoding of Categorical Data

Preparing Data

- Most of Machine learning algorithms can not handle categorical variables.
- We convert them to numerical values.

Nominal variable

One Hot Encoding

- Map each category to a vector that contains 1 and 0
 - 1 - presence of the feature
 - 0 - absence of the feature

Gender		isMale	isFemale	isOther
Male		1	0	0
Female		0	1	0
Other		0	0	1

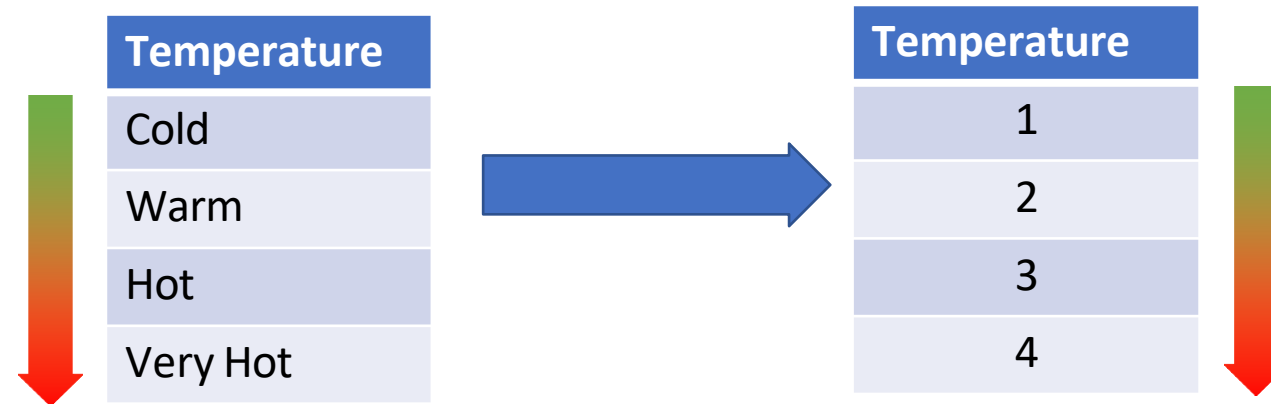
Encoding of Categorical Data

Preparing Data

Ordinal

Ordinal Encoding

- The encoding of variables retains the ordinal nature of the variable
- Each category is assigned a value from 1 through the number of possible values by considering the order of values.



Encoding of Categorical Data

Preparing Data

Practice

How can we encode the following categorical data?

Align
Bad
Neutral
Good

Hair
Black
Bronze
Brown
Gold
Gray

Normalization and Standardization

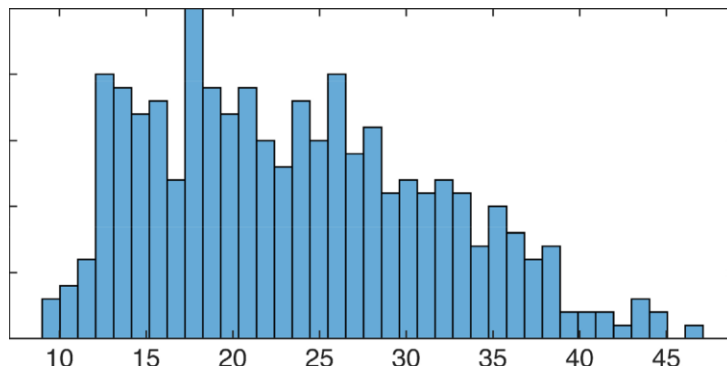
Preparing Data

Normalization

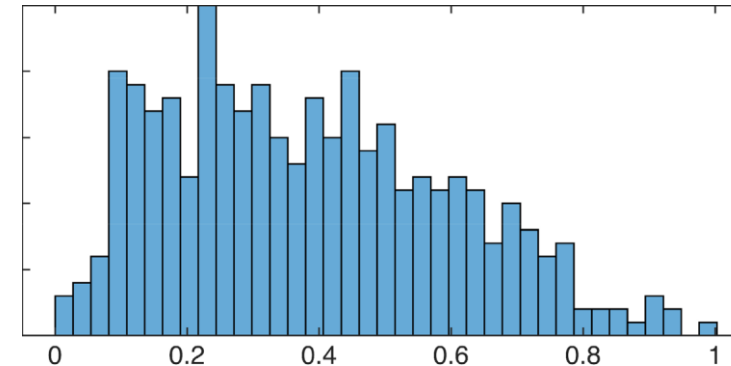
Scale a variable to have a values between 0 and 1

Min-Max Normalization:

$$x_{normalized} = \frac{x - x_{min}}{x_{max} - x_{min}}$$



Data distribution before normalized



Data distribution after normalized

Normalization and Standardization

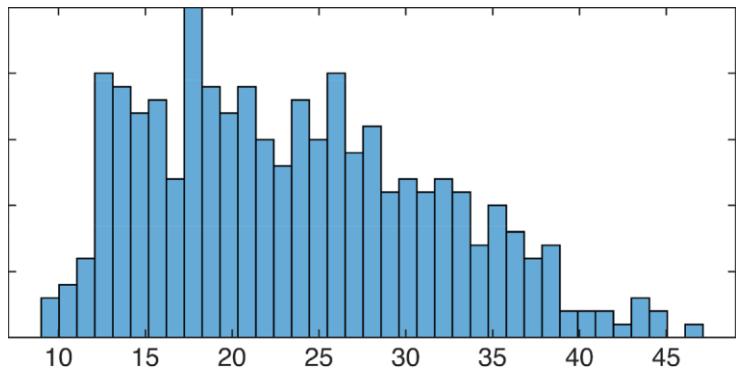
Preparing Data

Standardization

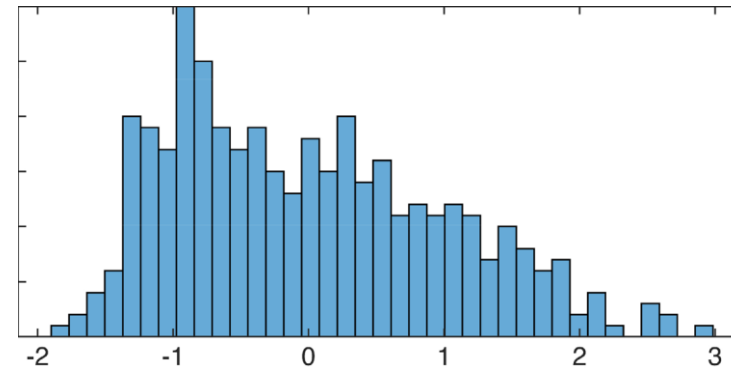
Transforms data to have a mean of zero and a standard deviation of 1.

Z-score Standardization

$$x_{standardized} = \frac{x - \bar{x}}{S.D.}$$



Data distribution before standardized



Data distribution after standardized

Data Quality

Preparing Data



Source:

<http://itsadeliverything.com/wordpress/images//accuracy-vs-precision.jpg>



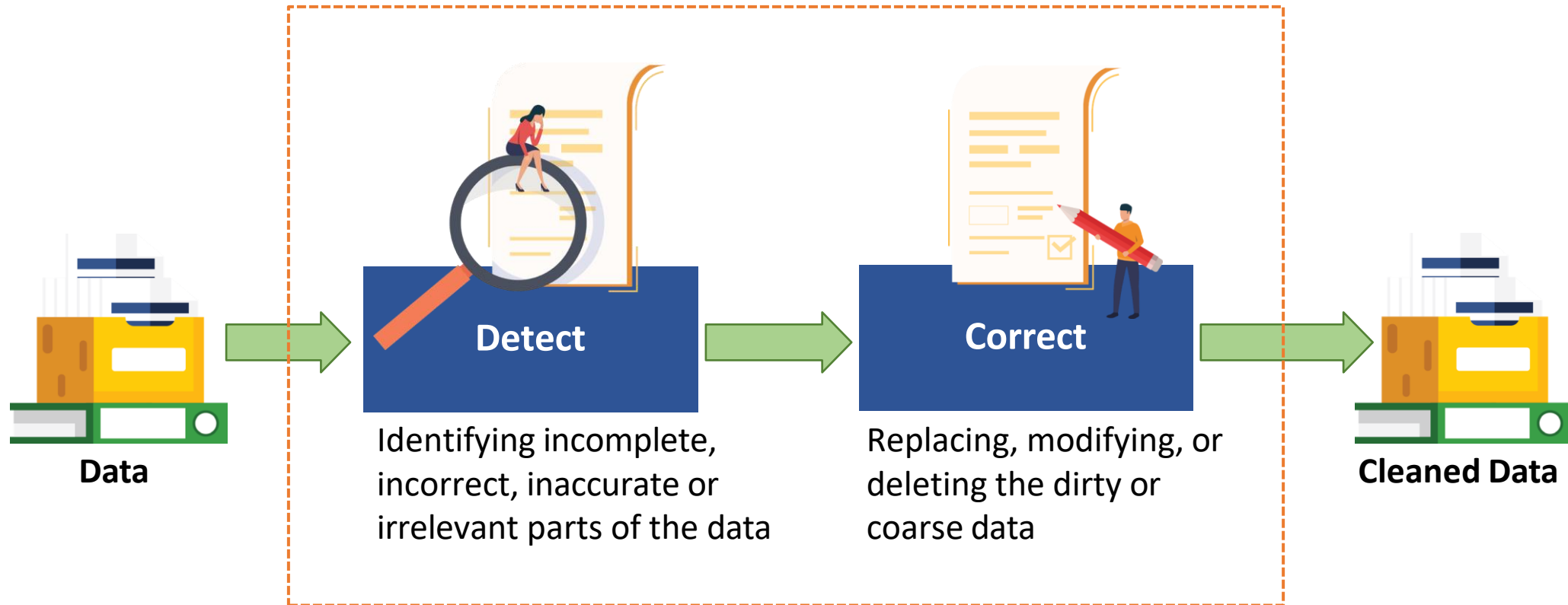
Data should be:

- **Accurate and Precise**
- **Complete** – Does not have "unknown" or "missing" values
- **Consistency** – Two data items in the data set contradict each other
- **Valid** – Conform to defined business rules or constraints
- **Uniform** – Using the same units of measure in all systems
- **Unique** – Does not contain duplicates

Data Cleaning

Preparing Data

Data Cleaning is the process of detecting and correcting/removing corrupt or inaccurate records from a record set



Inconsistent Datatypes

Preparing Data >> Data Cleaning

We expect that:

Values in a particular attribute must be of a particular datatype, e.g., Boolean, numeric (integer or real), date, etc.

Values in *align* and *alive* are inconsistent datatype

1 – Living Characters
0 – Deceased Characters

	name X_1	id X_2	align X_3	eye X_4	hair X_5	gender X_6	alive X_7	appearances X_8	first_appear X_9	publisher X_{10}
x_1	Spider-Man (Peter Parker)	Secret	Good	Hazel Eyes	Brown Hair	Male	1	4043	Aug-62	marvel
x_2	Captain America (Steven Rogers)	Public	Good	Blue Eyes	White Hair	Male	Living Characters	3360	Mar-41	marvel
	...	...	...	...	...	...	...	...	...	...
x_n	Natalia Romanova (Earth-616)	Public	1	Green Eyes	Red Hair	Female	Living Characters	1050	Apr-64	marvel

1 – Good
0 – Bad

Inconsistent Datatypes

Preparing Data >> Data Cleaning

How to address the Inconsistent datatypes

- Choose an appropriate datatype
- Transform values in another datatype into the selected datatype

Values in *align* and *alive* are inconsistent datatype

1 – Living Characters
0 – Deceased Characters

	name X_1	id X_2	align X_3	eye X_4	hair X_5	gender X_6	alive X_7	appearances X_8	first_appear X_9	publisher X_{10}
x_1	Spider-Man (Peter Parker)	Secret	Good	Hazel Eyes	Brown Hair	Male	Living Characters	4043	Aug-62	marvel
x_2	Captain America (Steven Rogers)	Public	Good	Blue Eyes	White Hair	Male	Living Characters	3360	Mar-41	marvel
	...	...	...	...	...	...	...	...	...	...
x_n	Natalia Romanova (Earth-616)	Public	Good	Green Eyes	Red Hair	Female	Living Characters	1050	Apr-64	marvel

1 – Good
0 – Bad

Missing Value

Preparing Data >> Data Cleaning

We expect that:

All required measures are known.

	IQ X_1	Job performance X_2
x_1	78	NA
x_2	84	NA
x_3	84	NA
x_4	85	NA
x_5	99	7
x_6	105	10
x_7	105	11
x_8	106	15
x_9	108	10
x_{10}	112	10
x_{11}	113	12
x_{12}	115	14
x_{13}	118	16
x_{14}	134	12

Job performances of x_1, x_2, x_3
and x_4 are unknown.
They are missing value.

Missing Value

Preparing Data >> Data Cleaning

How to deal with the missing value

Single Imputation: Generate a single replacement value for each missing data point.

- **Arithmetic Mean Imputation**
 - replaces missing values with mean of available values
- **Regression Imputation**
 - replaces missing values with predicted scores from a regression equation
- **Hot-deck Imputation**
 - A collection of techniques that impute the missing values with scores from “similar” datapoints, such as nearest neighbor hot-deck and last observation carried forward.
- **and etc.**

Missing Value

Preparing Data >> Data Cleaning

	X_1	X_2
x_1	78	11.70
x_2	84	11.70
x_3	84	11.70
x_4	85	11.70
x_5	99	7
x_6	105	10
x_7	105	11
x_8	106	15
x_9	108	10
x_{10}	112	10
x_{11}	113	12
x_{12}	115	14
x_{13}	118	16
x_{14}	134	12

Example of Arithmetic Mean Imputation

1. Compute the arithmetic mean of X_2 from available values
2. Replace the missing values of X_2 by the arithmetic mean

Mean = 11.70

Missing Value

Preparing Data >> Data Cleaning

	IQ XX_1	Job performance XX_2
x_1	78	7.529
x_2	84	8.267
x_3	84	8.267
x_4	85	8.390
x_5	99	7
x_6	105	10
x_7	105	11
x_8	106	15
x_9	108	10
x_{10}	112	10
x_{11}	113	12
x_{12}	115	14
x_{13}	118	16
x_{14}	134	12

$$JP = 0.123(78) + (-2.065) = 7.529$$

$$JP = 0.123(84) + (-2.065) = 8.267$$

$$JP = 0.123(84) + (-2.065) = 8.267$$

$$JP = 0.123(85) + (-2.065) = 8.390$$

Example of Regression Imputation

1. Estimate a set of regression equations
2. Generate predicted values for the incomplete variables
3. Fill in the missing values

$$JP = \beta_1(III) + \beta_0 = 0.123(III) + (-2.065)$$

incomplete variables complete variables

Duplicate Data

Preparing Data >> Data Cleaning

We expect that:

A data should appear on the dataset one time

	name X_1	id X_2	align X_3	eye X_4	hair X_5	gender X_6	alive X_7	appearances X_8	first_appear X_9	publisher X_{10}
x_1	Spider-Man (Peter Parker)	Secret	Good	Hazel Eyes	Brown Hair	Male	Living Characters	4043	Aug-62	marvel
x_2	Captain America (Steven Rogers)	Public	Good	Blue Eyes	White Hair	Male	Living Characters	3360	Mar-41	marvel
x_3	Spider-Man (Peter Parker)	Secret	Good	Hazel Eyes	Black Hair	Male	Living Characters	NA	Aug-62	marvel
	...	...	...	...	...	...	...	...	...	...
x_{100}	Natalia Romanova (Earth-616)	Public	Good	Green Eyes	Red Hair	Female	Living Characters	1050	Apr-64	marvel

**We have two recodes of Spider-Man. So, the two recodes are duplicate data
Moreover, one contradicts each other**

Duplicate Data

Preparing Data >> Data Cleaning

How to deal with the duplicate data

1. Select one recode that is up-to-date and accurate
2. Remove the others

	name X_1	id X_2	align X_3	eye X_4	hair X_5	gender X_6	alive X_7	appearances X_8	first_appear X_9	publisher X_{10}
x_1	Spider-Man (Peter Parker)	Secret	Good	Hazel Eyes	Brown Hair	Male	Living Characters	4043	Aug-62	marvel
x_2	Captain America (Steven Rogers)	Public	Good	Blue Eyes	White Hair	Male	Living Characters	3360	Mar-41	marvel
	...	...	...	...	...	...	...	...	...	...
x_{100}	Natalia Romanova (Earth-616)	Public	Good	Green Eyes	Red Hair	Female	Living Characters	1050	Apr-64	marvel

We have two recodes of Spider-Man. So, the two recodes are duplicate data

Materials

Demo

- <https://colab.research.google.com/drive/1k2nTRy7gwTu6aigNOpLDgWno6inWLK1c?usp=sharing>

Demo Dataset

- https://drive.google.com/file/d/1Uqe_uNrXYNEI1gCeOxEjimalnmnHD_0q/view?usp=sharing